SMA 2102/STA 2105: Calculus II Tutorial 16/02/15

1. The market research department for a supermarket chain has determined that for one store the marginal price $p'(x)$ at x tubes per week for a certain brand of tooth paste is given by $p'(x) = -0.015e^{-0.01x}$. Find the price demand equation if the weekly demand is 50 tubes when the price is Kf. 2.35. Find the weekly demand when the price of a tube is Kf.1.89.
2. Repeat question 1 if $p'(x) = 0.09e^{0.01x}$ if the supplier is willing to supply 100 tubes per week at a price of Kf. 1.65. How many tubes would the supplier be willing to supply at a price of Kf.1.98 each.
3. The marginal price for weekly demand of x bottles of baby shampoo in a drug store is given by $p'(x) = \frac{-600}{(3x+30)^2}$. Find the price demand equation if the weekly demand is 150 when the price of a bottle of shampoo is Kf. 4. What is the weekly demand when the price is Kf. 2.50.
4. If the marginal profit in Kf. Is given by $p'(t) = 2t - te^{-t}$. Find the total profit earned at five years if $p(0) = 0$.
5. An oil field is estimated to produce oil at a rate of $R(t)$ thousands barrels per month t months from now is given by $R(t) = 10te^{-t}$, $R(0) = 0$. Find the production in the next five years.
6. The monthly sales of a particular personal computer are expected to decline at a rate of $s'(t) = -4te^{0.1t}$ computers per month where t is the time in months and $s(t)$ is the number of computers sold each month. The company plans to stop manufacturing this computer when the sales per month reach 800 computers. If the monthly sales now $t = 0$ is 2000 computers. Find $s(t)$. How long to the nearest month will the company continue to produce these computers.
7. Explain what is meant by the term integration. Hence given that the weekly marginal revenue from x pairs of tennis shoes is given by $R'(x) = 40 - 0.02x + \frac{200}{x+1}$ and $R(0) = 0$ where $R(x)$ is the revenue function. Find the total revenue from the sales of 1000 pairs of shoes.

8. The monthly sales of a particular program are expected to decline at a rate given by $\frac{ds}{dt} = -25t^{2/3}$ programs per month, where t is time in months and s is the number of each program sold each month. Microsoft plans to stop manufacturing this program when the monthly sales reach 800 programs. Find $s(t)$ hence determine how long will the company continue to manufacture this program if the production now is 2000 programs
9. After a person takes a pill, the drug contained in the pill is assimilated into the blood stream. The rate of assimilation t minutes after taking the pill is given by $R'(t) = \frac{201n(t+1)}{(t+1)}$ find the total amount of the drug that is assimilated into the blood stream during the first 10 minutes after the pill is taken.
10. Express $\frac{x^4 - 2x^3 + 3x^2 - x + 3}{x^3 - 2x^2 + 3x}$ in terms of partial fractions. Hence prove that

$$\int \frac{x^4 - 2x^3 + 3x^2 - x + 3}{x^3 - 2x^2 + 3x} dx = \frac{x^2}{2} + \ln\left|\frac{x}{\sqrt{x^2 - 2x + 3}}\right| + c$$

11. Evaluate the following integrals

    i. $\int \frac{x^4 - x^3 - x - 1}{x^3 - x^2} .dx$

    ii. $\int \frac{dx}{\sqrt{20 + 8x - x^2}}$

    iii. $\int \frac{d\theta}{5 - 2\sin\theta}$

    iv. $\int \frac{dx}{\sqrt{9 - 49x^2}}$

    v. $\int 3x^2 e^{(x^3 - 2)}$ dx

    vi. $\int x^2 \cos x dx$

    vii. $\int \frac{10x - 2x^2}{(x-1)^2 (x+3)} dx$

    viii. $\int \frac{2x + 4}{(1 - x^2)(1 - x)} dx$

    ix. $\int \frac{dx}{\sin x - \cos x}$